

SET 2: CUSTOMER FEEDBACK ANALYSIS

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DATASET

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

QUESTION 1 – HISTOGRAM: DISTRIBUTION OF CUSTOMER AGES

AIM

To create a histogram in RStudio representing the distribution of customer ages.

ALGORITHM

1. Create a vector containing customer ages.
2. Use `hist()` to create the histogram.
3. Add labels and title.
4. Display the histogram in the Plots pane.

PROCEDURE IN RSTUDIO

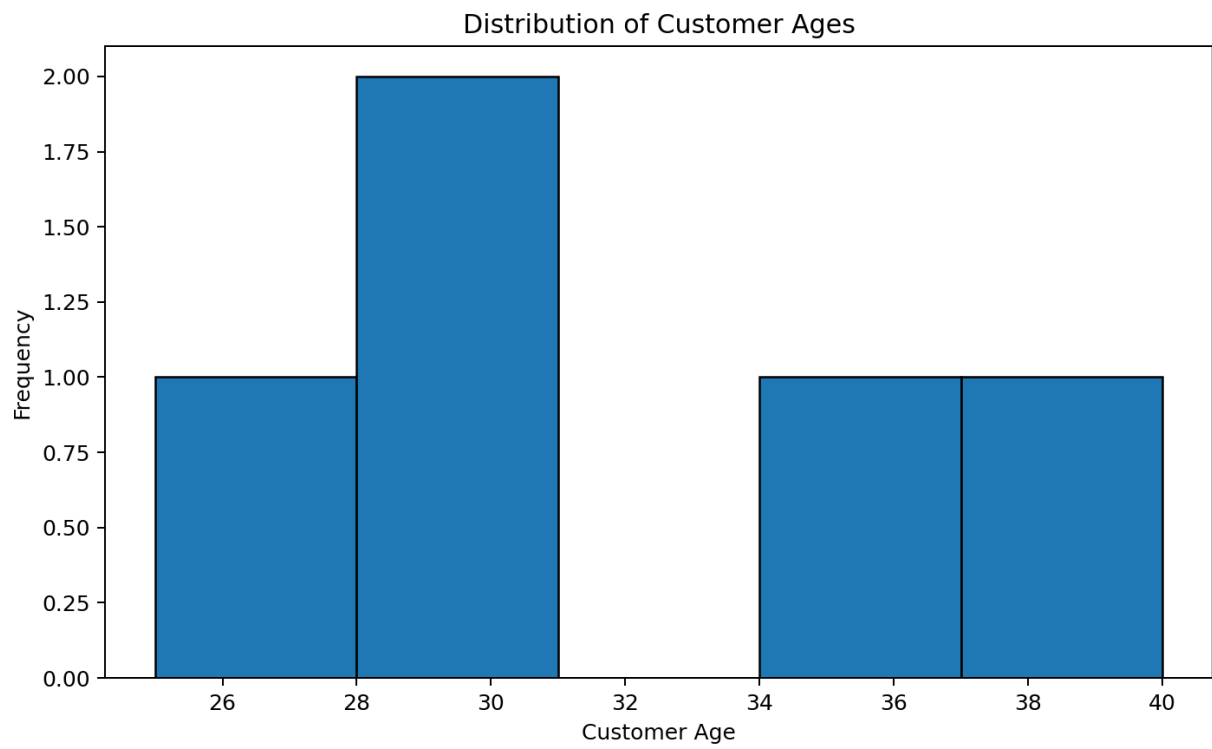
Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

R PROGRAMMING CODE FOR RSTUDIO

```
ages <- c(25, 30, 35, 28, 40)

hist(ages,
      breaks = 5,
      xlab = "Customer Age",
      ylab = "Frequency",
      main = "Distribution of Customer Ages",
      border = "black")
```

OUTPUT SCREENSHOT



RESULT

Thus, the customer age distribution was represented successfully using a histogram in RStudio.

QUESTION 2 – PIE CHART: CUSTOMER SATISFACTION SCORES

AIM

To generate a pie chart in RStudio showing the distribution of customer satisfaction scores.

ALGORITHM

1. Create the satisfaction score vector.
2. Count score frequencies using `table()`.
3. Calculate percentages.
4. Create labels.
5. Use `pie()` to display the chart.

PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

R PROGRAMMING CODE FOR RSTUDIO

```
scores <- c(4, 5, 3, 4, 5)

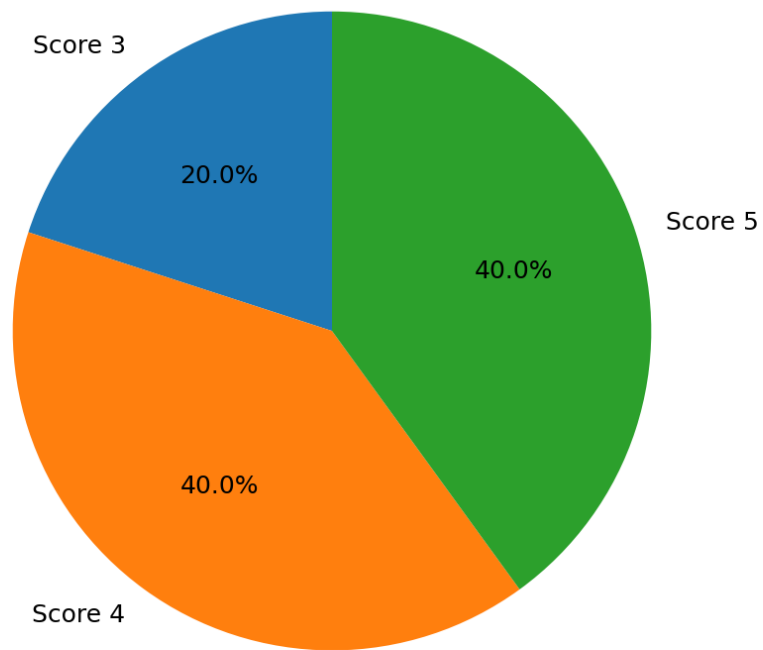
score_counts <- table(scores)
percentage <- round(100 * score_counts / sum(score_counts), 1)

pie_labels <- paste("Score", names(score_counts),
                    paste0("percentage, \"%")

pie(score_counts,
    labels = pie_labels,
    main = "Distribution of Customer Satisfaction Scores")
```

OUTPUT SCREENSHOT

Distribution of Customer Satisfaction Scores



RESULT

Thus, the customer satisfaction pie chart was generated successfully in RStudio. Scores 4 and 5 each represent 40% and score 3 represents 20%.

QUESTION 3 – STACKED BAR CHART: SATISFACTION BY AGE GROUP

AIM

To create a stacked bar chart in RStudio showing satisfaction score distribution by age group.

ALGORITHM

1. Create age and satisfaction score vectors.
2. Create age groups using `cut()`.
3. Generate a frequency table using `table()`.
4. Use `barplot()` to create stacked bars.
5. Add labels, title and legend.

PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

R PROGRAMMING CODE FOR RSTUDIO

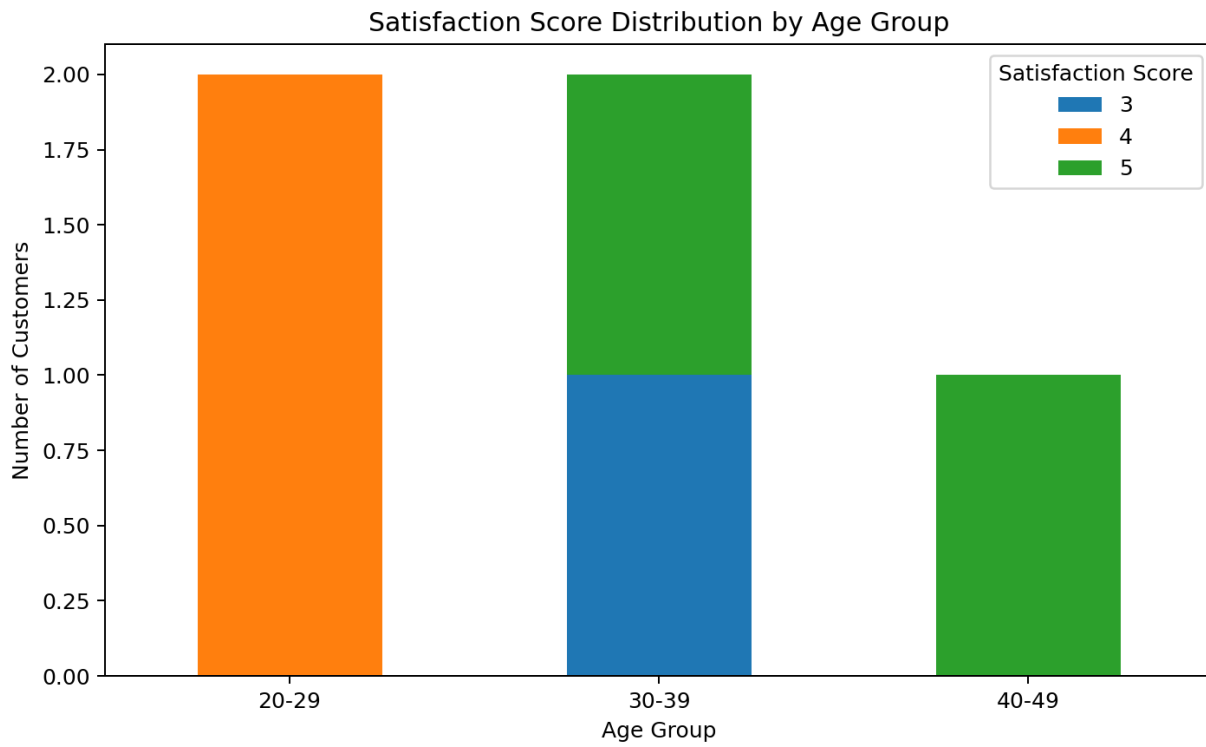
```
age <- c(25, 30, 35, 28, 40)
score <- c(4, 5, 3, 4, 5)

age_group <- cut(age,
                 breaks = c(20, 29, 39, 49),
                 labels = c("20-29", "30-39", "40-49"))

score_table <- table(score, age_group)

barplot(score_table,
        beside = FALSE,
        xlab = "Age Group",
        ylab = "Number of Customers",
        main = "Satisfaction Score Distribution by Age Group",
        legend.text = rownames(score_table))
```

OUTPUT SCREENSHOT



RESULT

Thus, the distribution of satisfaction scores by age group was visualized successfully using a stacked bar chart in RStudio.

QUESTION 4 – WORD CLOUD: CUSTOMER FEEDBACK

AIM

To develop a word cloud in RStudio from open-ended customer feedback and identify prevalent sentiments.

ALGORITHM

1. Install and load the wordcloud package.
2. Store the feedback text.
3. Split the feedback into individual words.
4. Calculate word frequencies.
5. Generate and display the word cloud.

PROCEDURE IN RSTUDIO

Open RStudio. Run `install.packages("wordcloud")` once in the Console. Create a new R Script, copy the program and click Run. The generated word cloud appears in the RStudio Plots pane. Larger words represent frequently occurring feedback terms.

R PROGRAMMING CODE FOR RSTUDIO

```
# Run this only once:
# install.packages("wordcloud")

library(wordcloud)

feedback <- paste(
  "excellent service very satisfied good experience friendly staff",
  "quick support happy customer quality service excellent product",
  "satisfied helpful staff good customer care easy process",
  "great experience fast response reliable service happy satisfied"
)

words <- unlist(strsplit(tolower(feedback), "\\s+"))
word_frequency <- sort(table(words), decreasing = TRUE)

wordcloud(words = names(word_frequency),
  freq = as.numeric(word_frequency),
  min.freq = 1,
  random.order = FALSE)
```

OUTPUT SCREENSHOT



RESULT

Thus, the customer feedback word cloud was generated successfully in RStudio. Frequently occurring positive words indicate predominantly positive customer sentiments.